THE UNIFIED QUANTUM & AI ROADMAP

MATH • ALGORITHMS • RESEARCH • 2026

PART I

Mathematical Foundations

Linear Algebra & Hilbert Spaces

Basis & Vectors

Machine learning maps data into high-dimensional feature vectors. Quantum computing extends this into complex Hilbert spaces.

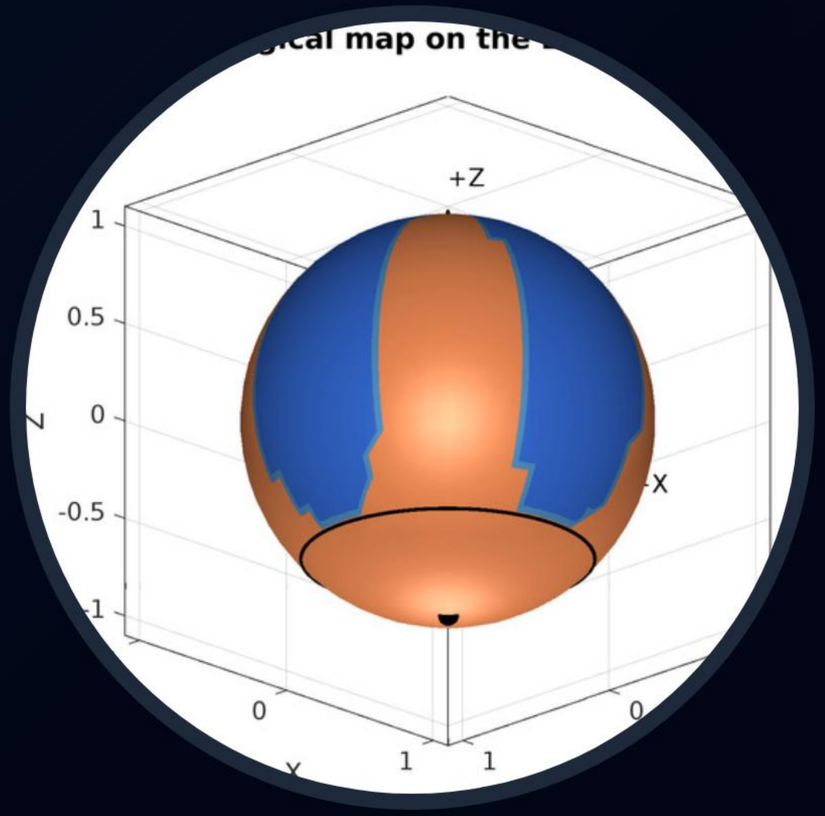
$$v = \sum_{i=0}^{n-1} c_i \, | i \rangle$$

Spectral Theory

Eigenvalues and Eigenvectors are critical for dimensionality reduction in ML and measurement observables in Quantum.

$$Ax = \lambda x$$

Probability & Uncertainty



The Logic of Randomness

Bayesian inference provides the logic for AI's probabilistic reasoning, while Born's Rule dictates the stochastic nature of quantum measurement.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Understanding entanglement requires a mastery of joint probability and entropy distributions.

PART II

AI & Machine Learning

Evolution of AI Algorithms

Symbolic AI

Early search and heuristic-based logic. Essential for constrained reasoning and deterministic planning.

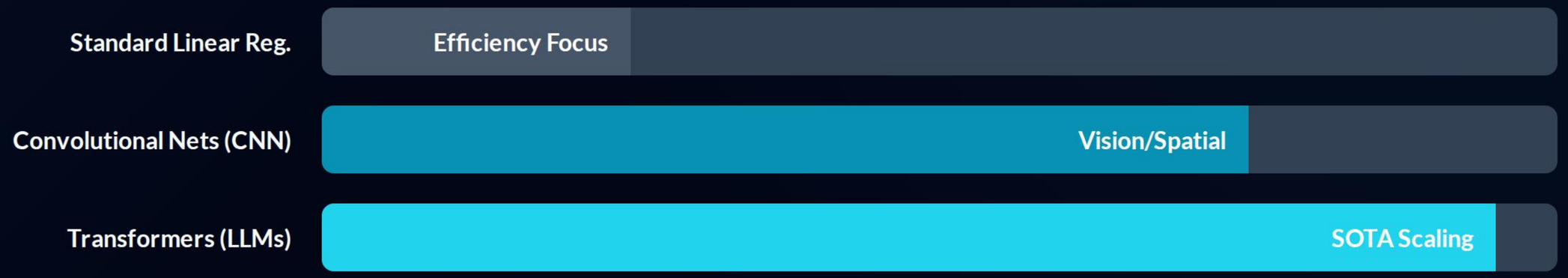
Statistical ML

Mapping inputs to outputs via data. SVMs, Random Forests, and Gradient Boosting for structured data.

Neural/GenAI

Deep Learning and Transformers. Leveraging massive compute to find patterns in unstructured data.

Algorithmic Performance Scaling



Scaling Laws: Model utility increases exponentially with parameter count and training compute.

PART III

The Quantum Frontier

Hardware Architectures

Scaling Qubits

Building a fault-tolerant quantum computer requires high fidelity and reliable error correction protocols.

- **Superconducting:** (IBM/Google) Fast gates but high noise, requires absolute zero.
- **Trapped Ions:** (IonQ/Quantinuum) High fidelity but slower execution speeds.
- **Photonic:** (Xanadu) Scalable architecture that operates at room temperature.

Quantum Dilution Refrigerator

Quantum x ML (QML)

Hybrid Architectures

QML leverages Variational Quantum Algorithms (VQA) where a classical optimizer trains a quantum circuit.

Core Merit: Quantum Kernels and Feature Maps can access high-dimensional spaces that are classically intractable, potentially solving complex chemistry and financial problems.

Parameter Shift Rule

$$\nabla f(\theta) = \frac{1}{2}\left(f\left(\theta + \frac{\pi}{2}\right) - f\left(\theta - \frac{\pi}{2}\right)\right)$$

Career Merit & Hierarchy

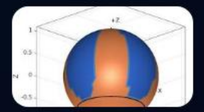
MERIT LEVEL	TYPICAL ROLE	REQUIRED MASTERY
Junior	ML Developer	Python, PyTorch, Linear Algebra
Senior	Senior AI Architect	Transformers, MLOps, Statistics
Principal	Research Scientist	Novel Kernels, Error Correction
Elite	QML Researcher	Hilbert Space, VQAs, Spin Physics

Questions?

Building the future at the intersection of bits and qubits.

roadmap.research.ai quantum-merit.io

Image Sources



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